

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-004 - Lakwa-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000040
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2000-12-27 / 2001-03-11
Measured / TVD	2526 m / 2485 m
Trajectory	Deviated; inc 11.6 deg; az 245.1 deg
Formation Interval	1506-2226 m
Lithology	Sandstone-siltstone sequence
Pore Pressure Reference	2716 psi
Fracture Gradient	14.26 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (45.1/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	631	Surface
9 5/8 in casing	1465	Intermediate
7 in liner/casing	2222	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2001-01-22 1003 m	Fishing Operation	High	BHA retrieval required specialized tools. Mitigation: Confirm fish geometry before selecting retrieval tool. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2001-01-29 1257 m	NPT	High	ROP dropped to zero during equipment unavailability. Mitigation: Maintain critical spares for recurring equipment failures. Lesson: Separate geological risk from equipment-driven NPT in analytics.
2001-02-01 1354 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2001-02-27 2142 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Avoid prolonged static periods in high-drag interval.
2001-03-03 2289 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2001-03-07 2399 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2001-01-05 430 m	Kick	Medium	Pit gain and flow with pumps off were observed. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Escalate connection-gas trends before entering correlated pressure zone.

Completion Summary

The synthetic well reached 2526 m MD in the Tipam interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.